

CSA0903 – PROGRAMMING IN JAVA

Class activity

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1. Write a program to store 15 integer values in an Array List and remove

```
5 write your code in this editor and press "Run" button to execute it.
6
7 *****/
8 import java.util.*;
9
10 public class Main {
11     public static void main(String[] args) {
12         ArrayList<Integer> a = new ArrayList<>(Arrays.asList(1,2,3,4,5,2,3,6,7,8,9,1,10,5,6));
13
14         for(int i=0;i<a.size();i++)
15             for(int j=i+1;j<a.size();j++)
16                 if(a.get(i).equals(a.get(j))) a.remove(j--);
17
18         System.out.println(a);
19     }
20 }
```

input

[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

...Program finished with exit code 0
Press ENTER to exit console.

2. Store employee IDs in a LinkedList and display them in both forward and reverse order.

```
7 *****/
8 import java.util.*;
9
10 public class Main{
11     public static void main(String[] args) {
12         LinkedList<Integer> l = new LinkedList<>(Arrays.asList(101,102,103,104,105));
13
14         System.out.println("Forward: " + l);
15
16         System.out.print("Reverse: ");
17         ListIterator<Integer> it = l.listIterator(l.size());
18         while(it.hasPrevious())
19             System.out.print(it.previous()+" ");
20     }
21 }
```

input

Forward: [101, 102, 103, 104, 105]
Reverse: 105 104 103 102 101

...Program finished with exit code 0
Press ENTER to exit console.

3. Store email IDs entered by the user in a HashSet. Display only unique email IDs and count the total unique entries.

```
7 *****
8 import java.util.*;
9
10 public class Main{
11     public static void main(String[] args) {
12         Scanner sc = new Scanner(System.in);
13         HashSet<String> h = new HashSet<>();
14
15         System.out.print("Enter number of emails: ");
16         int n = sc.nextInt();
17         sc.nextLine();
18
19         for(int i=0;i<n;i++)
20             h.add(sc.nextLine());
21
22         System.out.println("Unique Emails: " + h);
23         System.out.println("Count: " + h.size());
24     }
25 }
```

input

```
Enter number of emails: 5
abc@gmail.com
bcd@gmail.com
trs@gmail.com
mno@gmail.com
hij@gmail.com
Unique Emails: [mno@gmail.com, bcd@gmail.com, hij@gmail.com, trs@gmail.com, abc@gmail.com]
Count: 5

...Program finished with exit code 0
Press ENTER to exit console.
```

4. Write a Java program to count the frequency of each word in a paragraph using a HashMap.

```
6
7 *****/
8 import java.util.*;
9
10 public class Main{
11     public static void main(String[] args) {
12         Scanner sc = new Scanner(System.in);
13
14         System.out.println("Enter paragraph:");
15         String s = sc.nextLine();
16
17         HashMap<String,Integer> m = new HashMap<>();
18
19         for(String w : s.split(" ")) {
20             m.put(w, m.getOrDefault(w,0)+1);
21         }
22
23         System.out.println(m);
24     }
25 }
```

input

```
Enter paragraph:
ava is easy and java is powerful
java=2, powerful=1, and=1, is=2, easy=1}

..Program finished with exit code 0
Press ENTER to exit console.
```